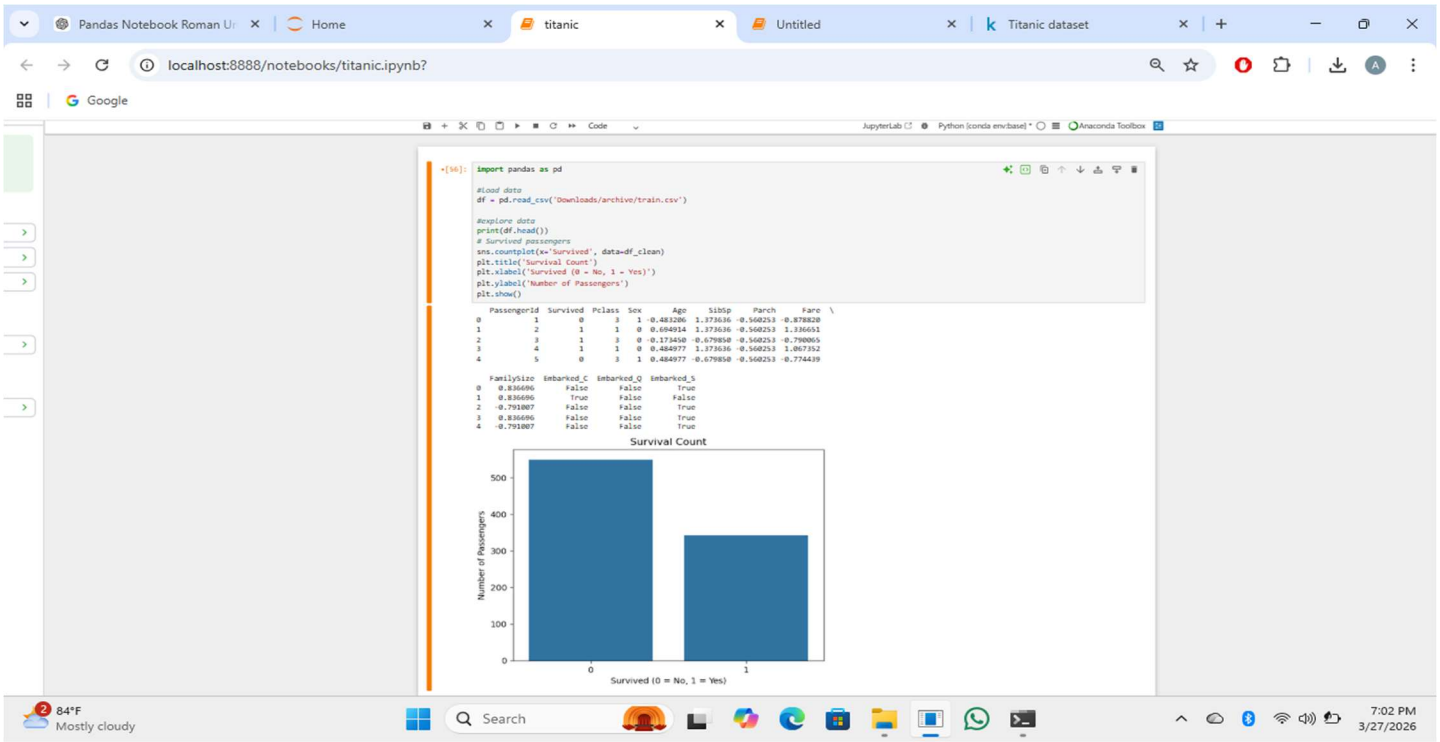
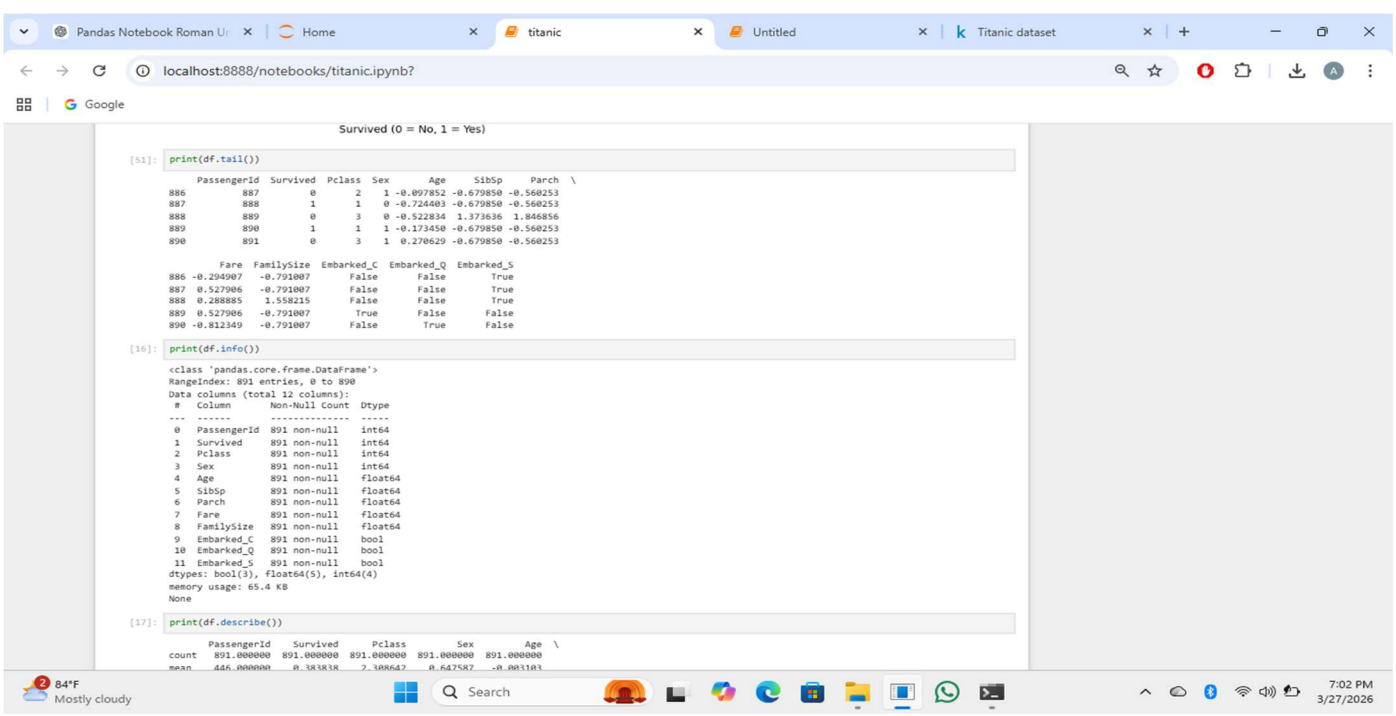
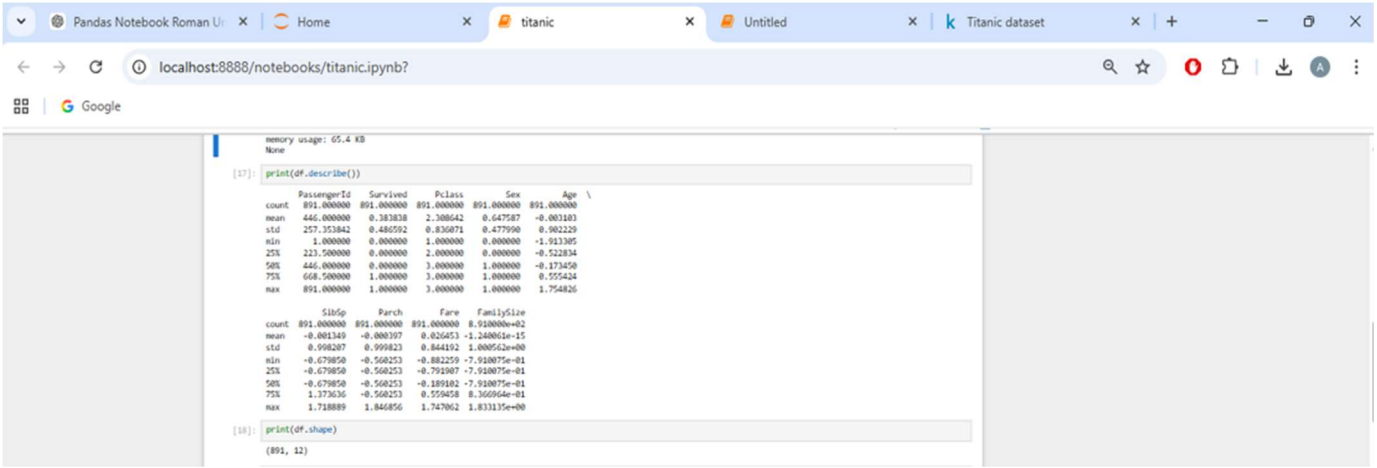
Practical task 4: Titanic dataset

Load data:

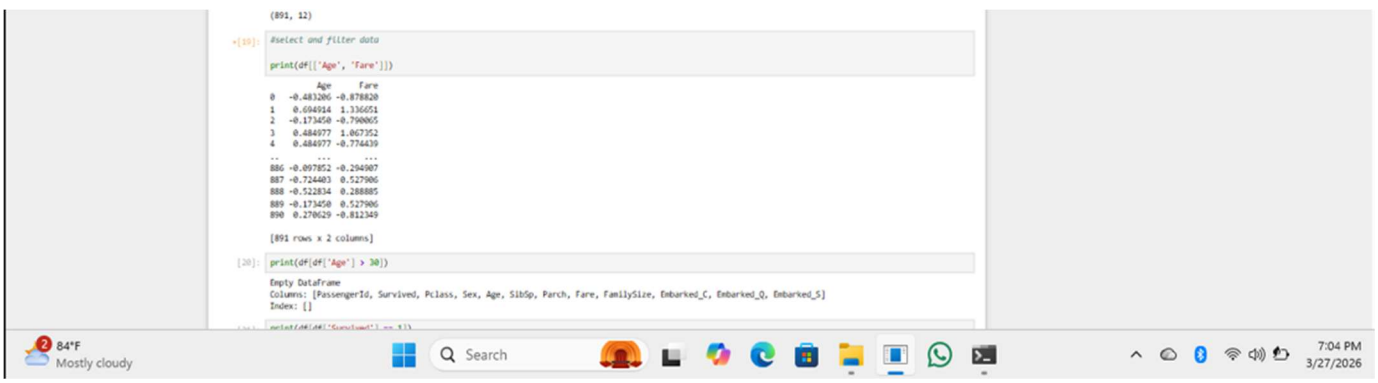


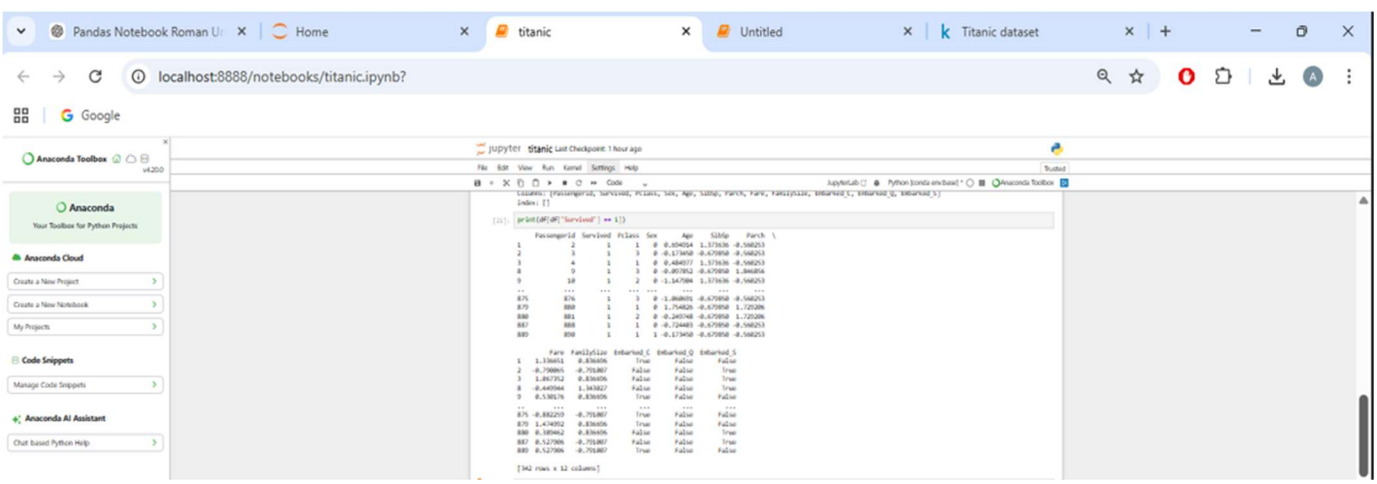
Explore data:





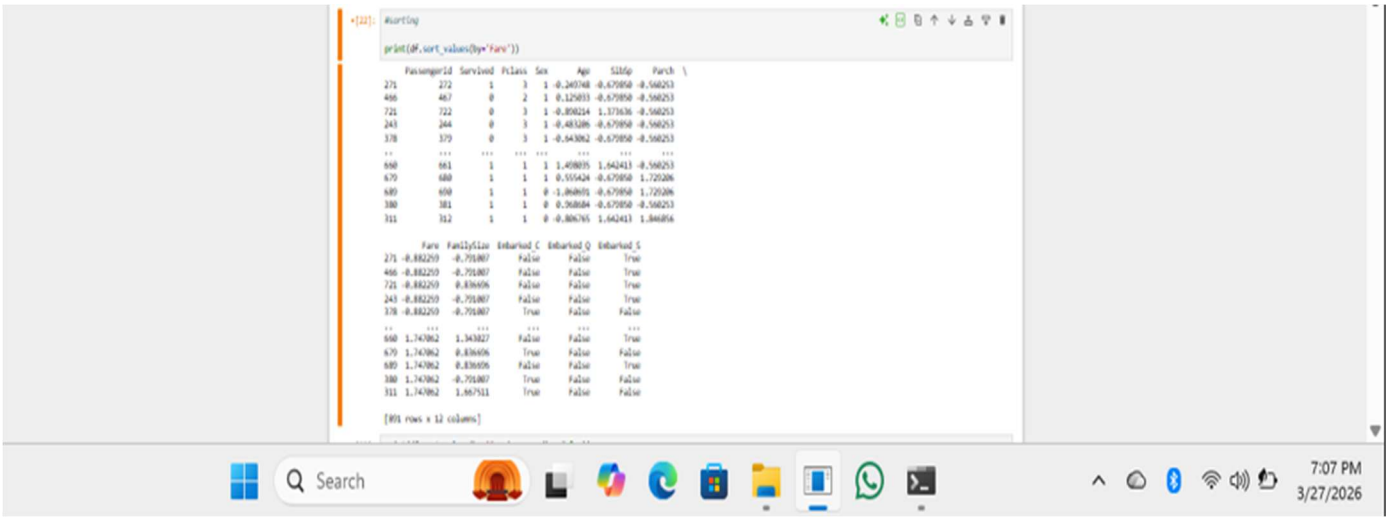
Select columns and filter rows:

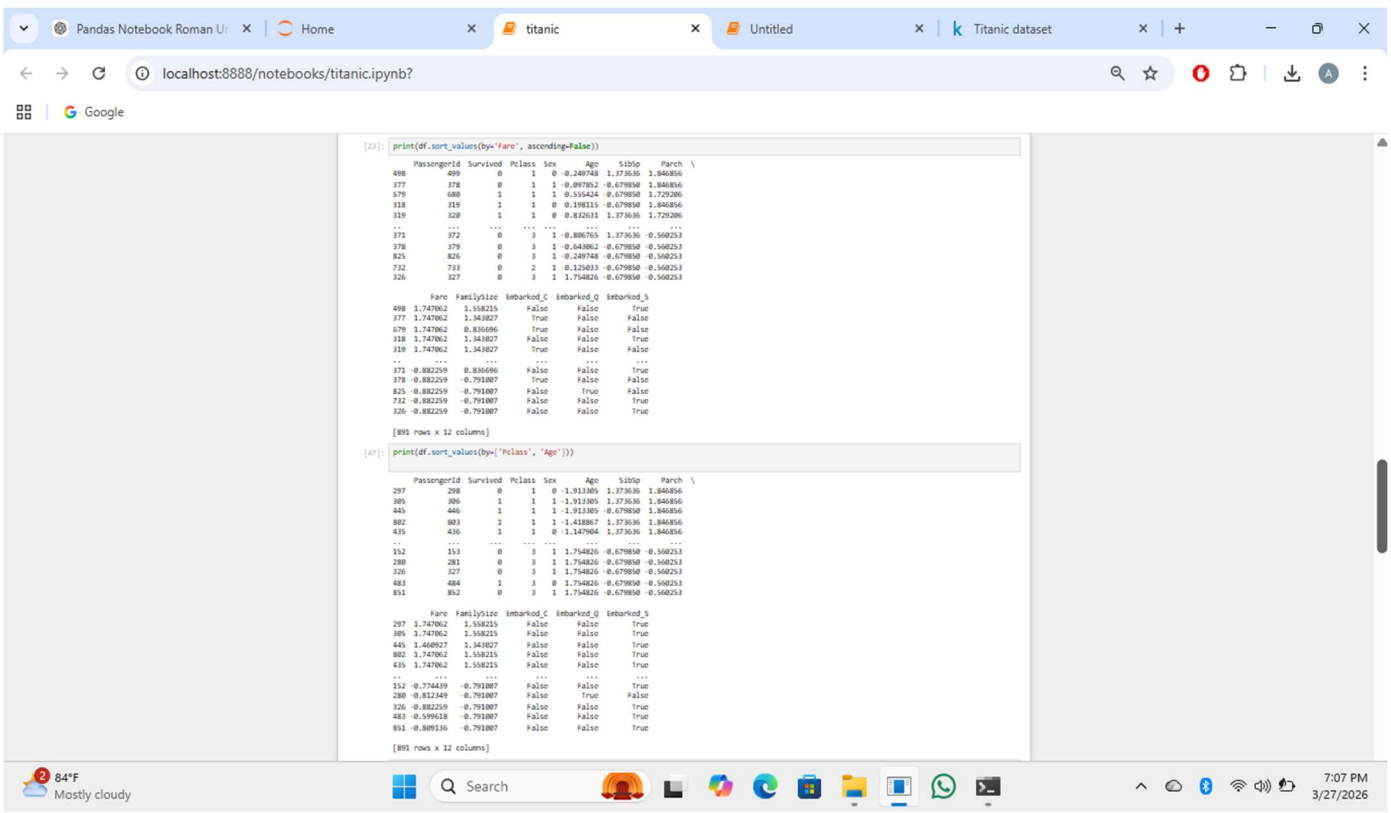




Sort:

```
[22]: #sorting

print(df.sort_values(by='Fare'))

     PassengerId  Survived  Pclass  Sex       Age    SibSp    Parch  \
271          272         1       3    1 -0.240748 -0.679850 -0.560253
466          467         0       2    1  0.125033 -0.679850 -0.560253
721          722         0       3    1 -0.850214  1.373636 -0.560253
243          244         0       3    1 -0.483286 -0.679850 -0.560253
378          379         0       3    1 -0.643062 -0.679850 -0.560253
..           ...       ...     ...  ...       ...       ...       ...
660          661         1       1    1  1.498035  1.642413 -0.560253
679          680         1       1    1  0.555424 -0.679850  1.729206
689          690         1       1    0 -1.860691 -0.679850  1.729206
380          381         1       1    0  0.968684 -0.679850 -0.560253
311          312         1       1    0 -0.806765  1.642413  1.846856

         Fare  FamilySize  Embarked_C  Embarked_Q  Embarked_S
271 -0.882259   -0.791007       False       False        True
466 -0.882259   -0.791007       False       False        True
721 -0.882259    0.836696       False       False        True
243 -0.882259   -0.791007       False       False        True
378 -0.882259   -0.791007        True       False       False
..        ...         ...         ...         ...         ...
660  1.747062    1.343027       False       False        True
679  1.747062    0.836696        True       False       False
689  1.747062    0.836696       False       False        True
380  1.747062   -0.791007        True       False       False
311  1.747062    1.667511        True       False       False

[891 rows x 12 columns]
```





Handle missing data:

```
[25]: #missing data handling

print(df.isnull().sum())
```

```
PassengerId    0
Survived       0
Pclass         0
Sex            0
Age            0
SibSp          0
Parch          0
Fare           0
FamilySize     0
Embarked_C     0
Embarked_Q     0
Embarked_S     0
dtype: int64
```

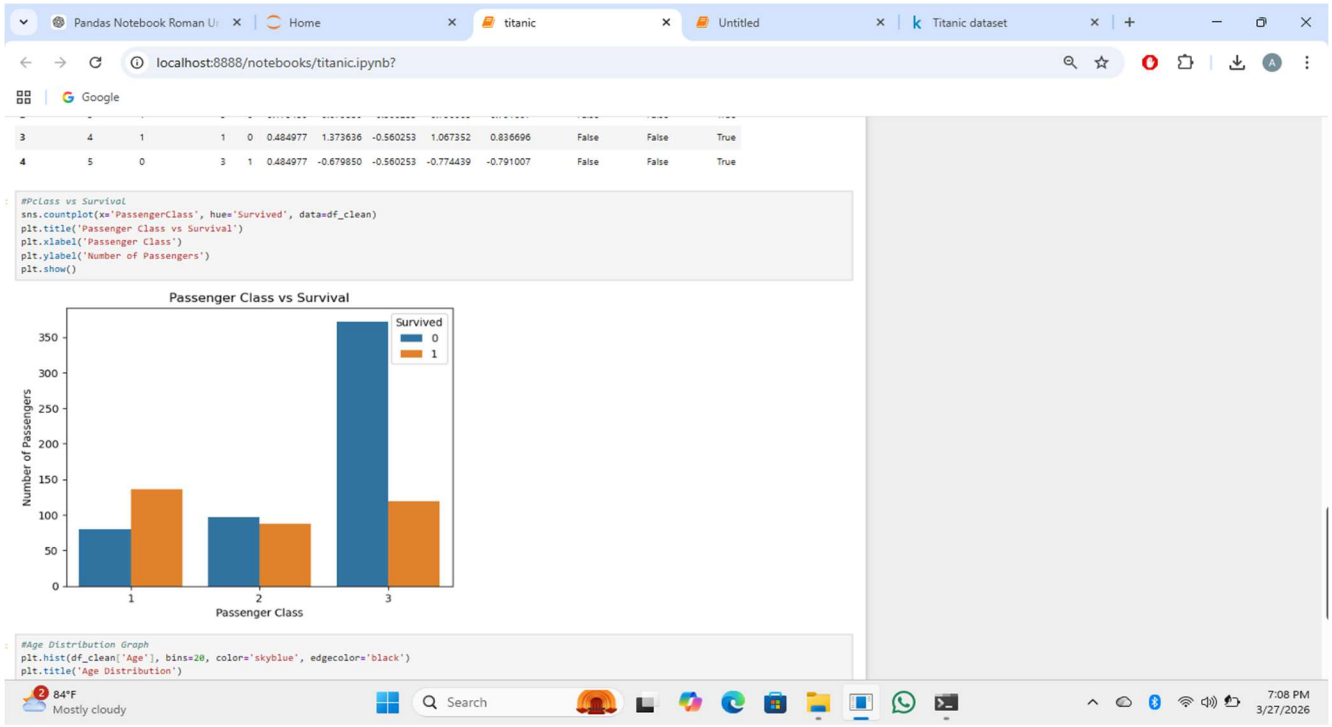
Add, remove, drop column:

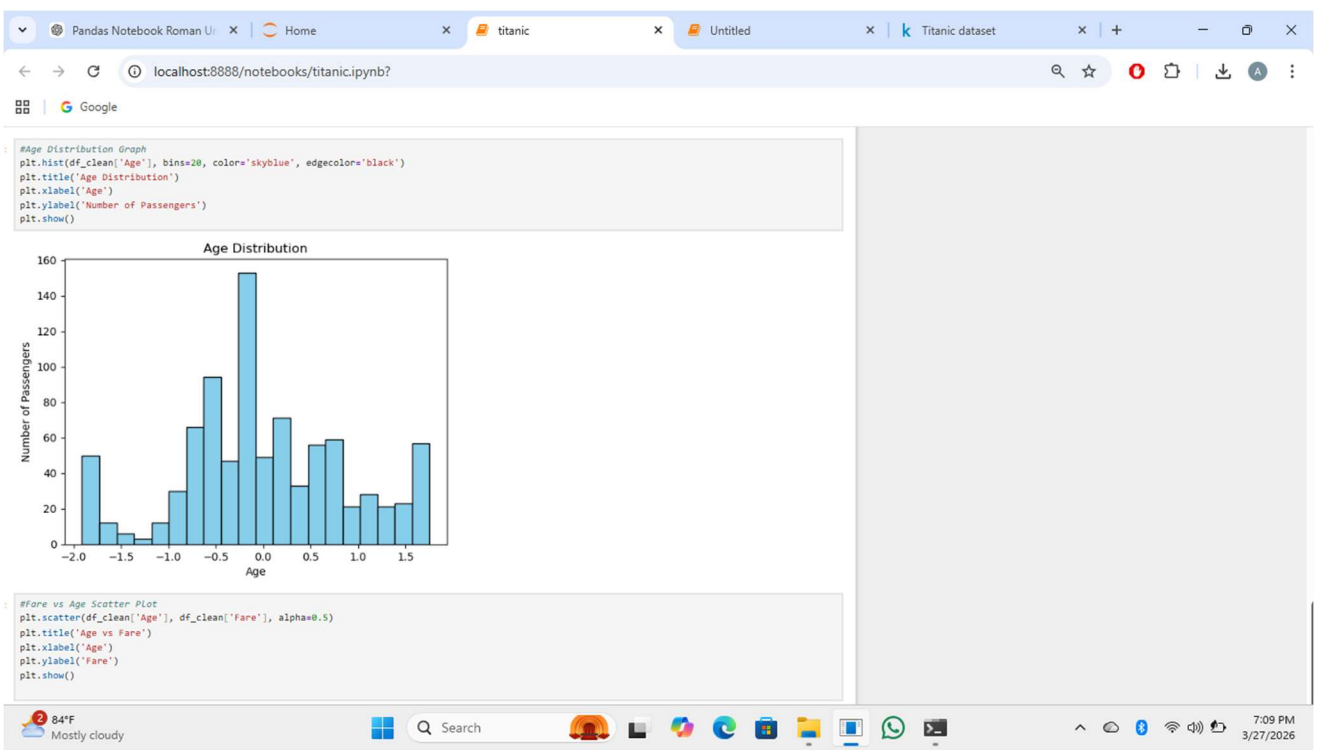
```
[48]: #coulmn operation
ndf = df.dropna()
df['Age'] = df['Age'].fillna(df['Age'].mean())
df_clean = df.dropna()
df['FamilySize'] = df['SibSp'] + df['Parch']
df_clean = df_clean.rename(columns={'Pclass':'PassengerClass'})
df_clean.head()
```

[48]:

	PassengerId	Survived	PassengerClass	Sex	Age	SibSp	Parch	Fare	FamilySize	Embarked_C	Embarked_Q	Embarked_S
0	1	0	3	1	-0.483206	1.373636	-0.560253	-0.878820	0.836696	False	False	True
1	2	1	1	0	0.694914	1.373636	-0.560253	1.336651	0.836696	True	False	False
2	3	1	3	0	-0.173450	-0.679850	-0.560253	-0.790065	-0.791007	False	False	True
3	4	1	1	0	0.484977	1.373636	-0.560253	1.067352	0.836696	False	False	True
4	5	0	3	1	0.484977	-0.679850	-0.560253	-0.774439	-0.791007	False	False	True

Graphs:


Pandas Notebook Roman U
Home
titanic
Untitled
Titanic dataset
localhost:8888/notebooks/titanic.ipynb?
Google
3 4 1 1 0 0.484977 1.373636 -0.560253 1.067352 0.836696 False False True
4 5 0 3 1 0.484977 -0.679850 -0.560253 -0.774439 -0.791007 False False True
#Pclass vs Survival
sns.countplot(x='PassengerClass', hue='Survived', data=df_clean)
plt.title('Passenger Class vs Survival')
plt.xlabel('Passenger Class')
plt.ylabel('Number of Passengers')
plt.show()
Passenger Class vs Survival
Survived
0
1
Number of Passengers
Passenger Class
#Age Distribution Graph
plt.hist(df_clean['Age'], bins=20, color='skyblue', edgecolor='black')
plt.title('Age Distribution')
84°F
Mostly cloudy
Search
7:08 PM
3/27/2026


Pandas Notebook Roman U
Home
titanic
Untitled
Titanic dataset
localhost:8888/notebooks/titanic.ipynb?
Google
#Age Distribution Graph
plt.hist(df_clean['Age'], bins=20, color='skyblue', edgecolor='black')
plt.title('Age Distribution')
plt.xlabel('Age')
plt.ylabel('Number of Passengers')
plt.show()
Age Distribution
Number of Passengers
Age
#Fare vs Age Scatter Plot
plt.scatter(df_clean['Age'], df_clean['Fare'], alpha=0.5)
plt.title('Age vs Fare')
plt.xlabel('Age')
plt.ylabel('Fare')
plt.show()
84°F
Mostly cloudy
Search
7:09 PM
3/27/2026